

Mitochondrial regulator prioritization

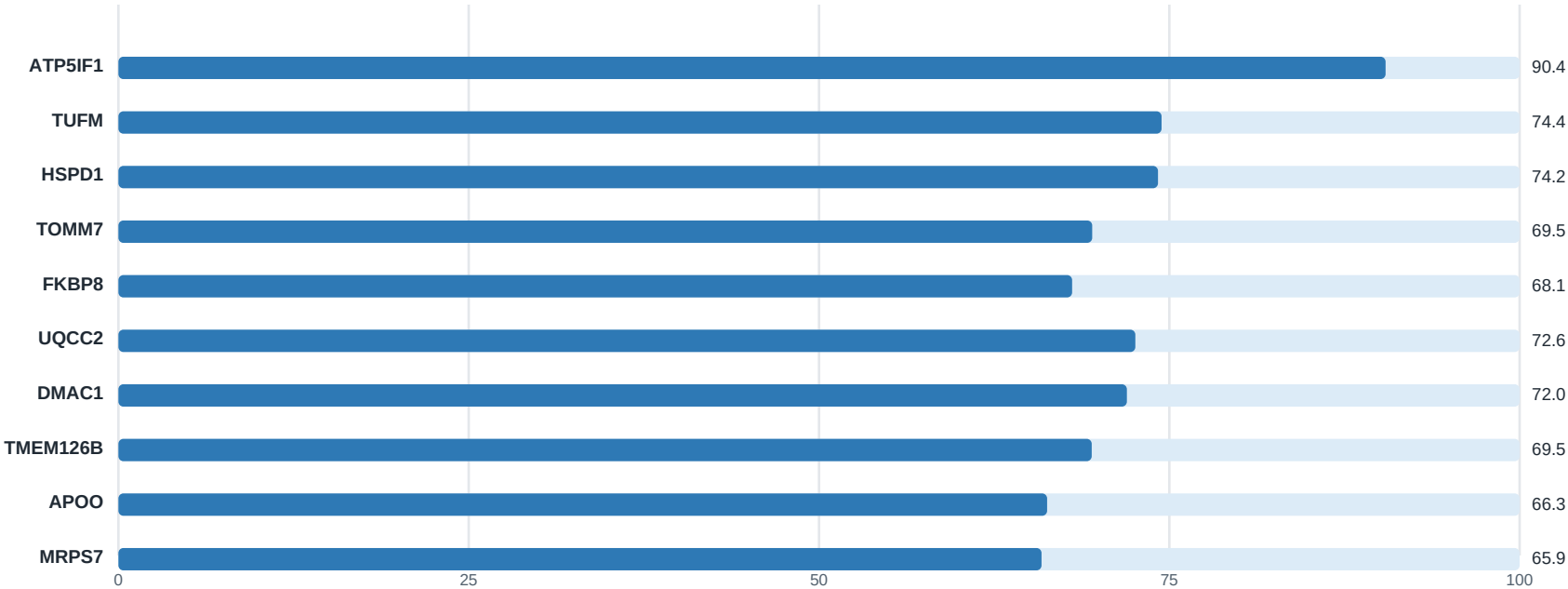
A first-pass ranking from mitochondrial DEGs, cross-stratum similarity tails, pathway enrichment, and curated mitochondrial control roles.

Interpretation boundary
The current data nominate regulators; they do not establish key drivers. A key-driver call requires directed neighborhood enrichment in the lab's cell-type Bayesian networks.

1,195 core-mito candidates	321 / 324 modeled AD-NCI contexts	0 gene-level similarity FDR hits	6 / 6 OXPHOS-enriched comparisons
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Local evidence score by regulatory-control candidate

Scores integrate DEG recurrence/effect, cross-context divergence, pathway-query support, and data coverage. They are ranks, not probabilities or p-values.



Source: Phase 09 DEGs + Phase 10 low-similarity ranks + Phase 11 ORA; production validated July 23, 2026.

What stands out

ATP5IF1	Strongest strict local candidate: 34 significant contexts, 8/9 broad lineages, all six divergent tails, and membership in every significant focused OXPHOS query.
TUFM TOMM7 FKBP8	Mechanistically diverse bridge from the OXPHOS phenotype to translation, PINK1/Parkin entry, and receptor-mediated mitophagy.
HSPD1	All 22 significant female-APOE4 contexts are down; median log2FC across all 54 modeled clusters is -0.89.
Shared reversal	Top candidates tend to be higher in female-APOE2 AD but lower in female-APOE4 and/or male-APOE2 AD.

Candidate-level evidence

Rank	Gene	Score	DE contexts	Lineages	Strata	Low tails	Control program / role
1	ATP5IF1	90.4	34	8/9	5/6	6/6	ATP synthase inhibitory factor subunit 1
2	TUFM	74.4	19	7/9	6/6	6/6	genome expression
3	HSPD1	74.2	37	9/9	6/6	4/6	proteostasis import
4	TOMM7	69.5	25	7/9	6/6	6/6	dynamics surveillance,proteostasis import
5	FKBP8	68.1	28	6/9	6/6	6/6	dynamics surveillance
6	UQCC2	72.6	12	6/9	4/6	5/6	oxphos assembly
7	DMAC1	72.0	12	4/9	5/6	5/6	oxphos assembly
8	TMEM126B	69.5	11	4/9	5/6	5/6	oxphos assembly
9	APOO	66.3	21	8/9	4/6	4/6	dynamics surveillance
10	MRPS7	65.9	13	5/9	5/6	6/6	genome expression
11	TIMM13	64.0	17	4/9	5/6	6/6	proteostasis import
12	FIS1	63.0	22	5/9	5/6	5/6	dynamics surveillance
13	SLIRP	61.3	20	7/9	3/6	5/6	genome expression
14	NDUFAF4	63.8	12	4/9	4/6	4/6	oxphos assembly
15	PARK7	62.9	21	7/9	4/6	3/6	dynamics surveillance,proteostasis import

Median AD-NCI effect by sex/APOE stratum

Cells show median log2 fold-change across every modeled fine cell type; positive values indicate AD higher than NCI.

	F-e2	F-e33	F-e4	M-e2	M-e33	M-e4
ATP5IF1	+0.10	+0.09	-0.45	-0.59	-0.17	-0.19
TUFM	+0.03	+0.02	-0.10	-0.29	-0.11	-0.20
HSPD1	+0.09	+0.03	-0.89	+0.11	-0.19	+0.00
TOMM7	+0.02	+0.18	-0.50	-0.60	-0.10	-0.14
FKBP8	+0.22	+0.13	-0.16	-0.44	+0.00	+0.07

AD higher AD lower

Source: Phase 09 MAST, all modeled fine-cell contexts; ROSMAP postmortem cohort.

Validation contrast

Start with female APOE4 excitatory neurons and astrocytes (26 AD, 11 NCI donors). Use female APOE2 as the reversal arm (8 AD, 17 NCI). Treat male APOE2 as replication because it has only 7 AD and 6 NCI donors.

Pathway phenotype is consistent and downstream-heavy

Every low-similarity comparison enriches MitoCarta OXPHOS subunits. This validates the phenotype but explains why structural ETC genes dominate naive rankings.

Comparison	Sig. MitoPathways	Top focused pathway	Top FDR	Fold	Sig. C2:CP
Sex: F vs M (all APOE)	8	OXPHOS subunits	9.4e-14	2.47x	64
APOE: e2 vs e33	8	OXPHOS subunits	5.4e-14	2.50x	62
APOE: e4 vs e33	6	OXPHOS subunits	1.2e-09	2.22x	35
Sex within e2	8	OXPHOS subunits	1.2e-16	2.66x	41
Sex within e33	7	OXPHOS subunits	6.6e-11	2.33x	35
Sex within e4	6	OXPHOS subunits	2.0e-08	2.13x	32

Source: Phase 11 core_mito bottom-200 ORA; MitoCarta 3.0 and MSigDB C2:CP 2026.1.

Recommended first experimental panel

Primary perturbation screen	ATP5IF1 TUFM TOMM7 FKBP8 HSPD1 Best combination of local recurrence, control-point biology, cross-stratum divergence, and external disease evidence.
Paired cristae mechanism	CHCHD2 CHCHD10 Both are locally strong in all six tails; perturb together because functional redundancy may mask single-gene effects.
Canonical pathway controls	PINK1 PHB2 Lower local rank but established mitophagy anchors for TOMM7/FKBP8 perturbations.
Molecular readouts	MT-ND2 COX4I1 COX5B ATP5F1E Strong downstream phenotype markers; measure them without calling them upstream drivers.

Suggested readouts: basal/maximal OCR, ATP, membrane potential, mitochondrial ROS, mitophagy flux, cristae morphology, and transcript/protein levels for MT-ND2 plus representative Complex I/IV/V subunits.

External evidence that changes priority

- ATP5IF1:** A 2026 human-brain study reports hippocampal loss of protective IF1 in sporadic AD. [Source](#)
- TUFM:** Loss links mitochondrial ROS to BACE1/A-beta, apoptosis, and tau phosphorylation. [Source](#)
- FKBP8:** A 2024 study connects phospho-tau to impaired FKBP8-mediated mitophagy. [Source](#)
- TOMM7:** A genome-wide screen identifies TOMM7 as required for PINK1 stabilization and Parkin recruitment. [Source](#)
- HSPD1:** Human AD mtUPR data support disease involvement; the local female-e4 direction suggests loss of compensation. [Source](#)

Decision gate for key driver

1. Map the 54 fine clusters to each available cell-type Bayesian network.
2. Project pathway- and contrast-specific mitochondrial DEG signatures.
3. Test directed 1-3 hop neighborhoods and BH-correct across candidate hubs.
4. Require replication in coexpression modules and donor-level pseudobulk.
5. Add AD GWAS/eQTL/TWAS evidence as an orthogonal score, not a substitute for KDA.

Keep the label strict

Until this gate is passed, use "candidate regulator" or "pre-network priority," not "key driver."

Reproducible sources

- Canvas:** mitochondrial-regulator-prioritization.canvas.tsx
- Script:** scripts/analysis/mt_pathway/prioritize_mitochondrial_regulators.py
- Score tables:** docs/analysis/mt_pathway/